Assignment Questions:

1. **Find 10 different model providers and their models released in last 12 months**

Model Provider	Model	Release Date	Cut-off Date
OpenAI	GPT-5.4	05-Mar-2026	31-Aug-2025
Google DeepMind	Gemini 3.1 Pro	19-Feb-2026	Jan-2025
Anthropic	Claude Sonnet 4.6	17-Feb-2026	Aug-2025
Alibaba Cloud (Qwen)	Qwen3.5 Flash	2026; exact date not verified	Not published
Z.ai	GLM-5	Feb-2026; exact date not verified	Not published
MiniMax	MiniMax-M2.5	Feb-2026; exact date not verified	Not published
DeepSeek	DeepSeek-V3.2	01-Dec-2025	Mar-2025
xAI	Grok 4.3	2026; exact date not verified	Not published
Moonshot AI	Kimi K2.5	27-Jan-2026	Not published
Mistral AI	Mistral Large 4	06-Oct-2026	Not published

Model	Cost of input tokens / 1M	Cost of output tokens / 1M	Context Window	Type	When to use this model
GPT-5.4	$2.50	$15.00	1.05M	Reasoning, coding, multimodal, agentic	Complex problem-solving, software development, spreadsheets, presentations and computer-use agents
Gemini 3.1 Pro	$2.00*	$12.00*	1M	Multimodal reasoning	Large documents, video, images, research, complex analysis and coding
Claude Sonnet 4.6	$3.00	$15.00	1M	General-purpose,	Production coding, document analysis,

				reasoning, coding	business workflows and agents
Qwen3.5 Flash	$0.029	$0.287	1M	General-purpose, multimodal	High-volume applications, fast responses, translation and data processing
GLM-5	Not independently verified	Not independently verified	Not independently verified	Reasoning, coding, agents	Complex coding, technical reasoning and agent workflows
MiniMax-M2.5	$0.304	$1.213	204,800	Coding, reasoning, agents	Programming, tool calling, office automation and productivity
DeepSeek-V3.2	$0.28–$0.56**	$0.42–$0.84**	164K	Reasoning, coding, agentic	Cost-conscious reasoning, programming, mathematical tasks and tool use
Grok 4.3	$1.25	$2.50	1M	Reasoning, general-purpose	Complex queries, long-context analysis and agent applications
Kimi K2.5	$0.574	$3.011	262,144	Multimodal, reasoning, coding	Document analysis, visual understanding, coding and agent tasks
Mistral Large 4	$1.36	$4.18	1M	Open-weight, multimodal, general-purpose	Enterprise AI, long-context applications, coding and customized deployments

2. Read 2–3 articles about how transformer works?

Goal to understand:

(a) Self-attention mechanism

(b) Vector embedding

(c) Feed forward Nueral Network

Answer:

A Transformer is the technology behind AI models like ChatGPT, Claude and Gemini. It helps computers understand language and generate answers.

There are 3 important concepts:

1. Vector Embedding – Converting words into numbers

Computers understand numbers, not words.

- The word "Cat" is converted into a list of numbers (vector).
- The word "Dog" gets another list of numbers.
- Words with similar meanings often have similar numerical representations.

Think of it as: Converting language into a format computers can understand.

2. Self-Attention – Understanding relationships between words

Self-attention helps AI understand which words are related to each other.

Example:

"The cat drank milk because it was hungry."

- AI identifies the word "it".
- It looks at other words in the sentence.
- It learns that "it" refers to "cat".
- Think of it as: Understanding the context and relationships between words.

3. Feed-Forward Neural Network – Processing information

After understanding relationships between words, the model processes the information using neural network layers.

- Takes the information from self-attention.
- Processes it through mathematical operations.
- Produces improved representations for the next layer.

Think of it as: A brain that processes the information it has gathered.